SPE-229587-MS

Beyond Limits: Delivering Over 1,000,000 Feet of Record-Setting Unconventional Wells in the UAE

M. Nour, E. Albeltagy, R. Bayov, A. Hebish, M. Anwar, A. T. Alajami, and E. D. Al Shamisi, ADNOC Onshore, Abu Dhabi, UAE

 DOI 10.2118/229587-MS

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This paper captures the transformational journey of ADNOC's unconventional (UC) oil drilling fleet over the past two years, during which over one million feet were safely drilled across multiple rigs. It showcases how strategic operational, engineering, and leadership initiatives systematically reduced well delivery time and cost, while maintaining outstanding safety performance. Beyond technical success, the campaign ignited a performance-driven culture that reshaped the future of UAE unconventional drilling, delivering consistent results even under challenging conditions.

A structured performance improvement framework was deployed, emphasizing empowered rig leadership, agile engineering support, and a daily focus on critical KPIs. Aggressive yet controlled drilling parameters were combined with optimized BHA designs and advanced drill bits to enhance penetration rates while maintaining borehole quality. Offline activities such as logging and wellhead works were systematically shifted off rig critical path. Rig teams embedded a culture of accountability around ROP, connection times, and flat time optimization. The campaign demanded adaptability, with drilling strategies continuously fine-tuned to overcome tight formations and elevated bottomhole temperatures exceeding 275°F, common across the UAE unconventional fields.

Across more than 82 wells, the fleet demonstrated outstanding gains:

- ROP peaked at 178.26 FPH in 17-1/2″, 137.39 FPH in 12-1/4″, and 124.25 FPH in 8-1/2″ sections
- A highest daily footage of 3,580 ft in 24 hours was recorded.
- Average well delivery time dropped by 80% from 96.5 days (Q3-2023) to 18.95 days (Q2-2025), with several wells touching the 14 days bar.
- Average cost per foot was reduced from $668/ft to $429/ft, achieving a 35.7% cost improvement.
- Weight-to-weight connection times were optimized with 60% improvement, significantly improving rig rhythm and cumulative operational efficiency.
- ILT sources were systematically targeted and minimized.
- Diesel consumption and related emission significantly decreased by more than 40%.

Notably, these improvements were realized while maintaining tool reliability and enhancing borehole quality, despite the added complexity of tight, abrasive formations and extreme temperature zones.

In addition to setting operational records, the campaign delivered measurable sustainability benefits. By saving hundreds of rig days, the fleet substantially reduced diesel consumption and CO_2 emissions, aligning with ADNOC's broader environmental objectives. The structured performance model, reinforced by strong leadership and adaptive engineering, provides a replicable and scalable blueprint for operational excellence in future unconventional drilling operations.

Introduction

Strategic Context and Motivation

The global energy landscape is undergoing a significant transformation, with an increasing emphasis on diversifying energy sources and enhancing energy security. Unconventional hydrocarbon resources are playing an increasingly pivotal role in meeting the world's growing energy demand. These resources require advanced technologies to unlock their potential.

The United Arab Emirates (UAE) is strategically positioned within this global context, possessing substantial unconventional reserves. The sheer scale of these reserves is not merely a geological fact; it serves as a powerful driving force behind ADNOC Onshore aggressive pursuit of efficiency and technological innovation.

ADNOC Onshore's UC drilling team has articulated a clear and resolute commitment to unlocking the UAE's vast unconventional reserves, recognizing their strategic importance for energy security. This commitment extends beyond operational metrics—it aligns with ADNOC's broader mandates for digital transformation and sustainability. The inherent complexities of drilling in the UAE's unconventional plays presented a formidable challenge. These reservoirs are characterized by tight formations and elevated bottomhole temperatures often exceeding 275°F, demanding highly specialized drilling techniques and robust equipment. Through innovation and strategic partnerships, ADNOC Onshore is driving efficiency and accelerating well deliveries. The achievements presented in this paper—reductions in time, cost, and emissions—are a direct result of this top-down strategic alignment that integrates operational excellence with broader corporate responsibility.

Prior to the transformational journey, unconventional drilling operations in the UAE faced substantial challenges that limited development pace and economic returns. Wells typically required months to complete, with costs exceeding $650 per foot—significantly higher than global benchmarks. Technical challenges included managing drilling dynamics in abrasive formations, maintaining downhole tool reliability at elevated temperatures, and achieving consistent borehole quality across heterogeneous lithologies. These constraints not only impacted economic returns but also limited the scalability required to fully capitalize on the resource potential.

Project Scope and Objectives

In response to these challenges, ADNOC Onshore UC drilling team launched an ambitious transformation journey, with the explicit goal of revolutionizing drilling performance while maintaining unwavering commitment to safety and environmental stewardship. The program established aggressive yet achievable targets: reducing well delivery time by 60%, decreasing cost per foot by over 25%, and drilling one million feet within a two-year timeframe—all while enhancing borehole quality and minimizing environmental footprint.

This paper serves as a comprehensive case study of how technical innovations, operational improvements, and cultural transformation converged to break long-standing performance barriers in one of the world's most demanding unconventional environments.

Strategic Framework for Performance Transformation: A Holistic Approach

The performance transformation that enabled ADNOC's UC drilling fleet to surpass one million feet drilled in record time was not the result of a single innovation, but the product of a purpose-built framework that integrated strategy, engineering, and execution into one cohesive system. This section outlines the three foundational pillars that underpinned the transformation: (1) a management-led performance system that aligned vision, accountability, and culture; (2) a robust technical strategy that translated subsurface complexity into consistent, high-speed drilling performance; and (3) an operational execution model that institutionalized discipline, rhythm, and responsiveness at the rig site. By weaving leadership intent, technical precision, and frontline momentum into a unified architecture, the campaign demonstrated how holistic integration—rather than isolated excellence—can unlock breakthrough performance at industrial scale.

Strategic Performance Management: Setting the Foundation for Transformation

The transformation was built on a tiered organizational structure that ensured seamless coordination across all drilling levels. Rig-site teams focused on daily operational excellence, field teams drove cross-rig optimization and technical support, while central leadership set direction, sustained momentum, and enabled knowledge transfer. A structured governance model supported this alignment, featuring daily performance updates, twice-weekly virtual reviews, weekly HSE meetings, and regular leadership rig visits. This tiered structure ensured timely escalation of challenges while maintaining unified focus on safety and performance objectives across the fleet. The model's success rested on four core enablers each instrumental in embedding a culture of high performance across the fleet. These are expanded in the following subsections.

Team Empowerment and Performance Culture. Central to the transformation was the deployment of an empowered rig leadership model that delegated day-to-day decision-making authority closer to the point of execution. Traditional hierarchies were replaced with a team-based system centered around the drilling supervisor, who was granted the autonomy to make operational decisions within a clearly defined accountability framework. Leadership capabilities were strengthened through fleet-wide performance reviews, structured coaching, and continuous communication of lessons learned. Performance expectations were clearly defined through a balanced scorecard capturing safety, quality, delivery, and cost indicators. This model enabled rapid responses to real-time conditions, accelerated learning, and embedded a culture of continuous improvement.

At the rig site, every crew member was encouraged to take ownership of performance targets. For instance, once trained in optimized connection practices, crews began suggesting refinements. This hands-on engagement fostered alignment, confidence, and agility—ensuring that every individual contributed meaningfully to operational success.

A key driver of the campaign's momentum was the intentional cultivation of a high-performance culture. Leadership instilled a mindset that "current success is not enough," encouraging teams to chase ambitious targets in ROP, connection time, and flat-time efficiency. Structured KPI reviews, intra-rig benchmarking, and frequent recognition—ranging from daily shout-outs to rig-of-the-month awards—fueled pride, peer-driven motivation, and a continuous push beyond previous limits.

Integrated KPI Governance and Support Structure. A key pillar of the transformation was the establishment of a performance governance system that tightly integrated KPI monitoring with leadership support and structured incentives. Rig teams operated within a daily performance loop built around clear KPIs—ROP, connection time, flat time percentage, NPT, and ILT hours. These metrics were reviewed twice daily through structured updates and benchmarked in twice-weekly virtual meetings, where teams jointly analyzed performance trends, upcoming plans, and HSE alignment. This sustained focus created a high-accountability environment where deviations were caught early and addressed before impacting operations.

To support this system, ADNOC Onshore UC leadership played an enabling role by providing resources, expediting policy changes, and ensuring top-down alignment. Daily KPI dashboards—visible at rig sites—offered real-time comparison of actual performance versus targets, fostering transparency and immediate course correction. Management reinforced this setup through weekly HSE meetings, regular rig visits, and a multi-tiered recognition program. Top-performing rigs were publicly acknowledged, and crew contributions were spotlighted through on-the-spot praise and structured reward programs. Financial incentives were introduced for contractor crews, directly linking frontline execution to measurable outcomes.

This integrated system of measurement, action, and recognition not only sustained performance momentum but also synchronized the entire drilling ecosystem into a culture of shared accountability and continuous improvement.

Collaboration, Knowledge Sharing, and Continuous Learning. A key enabler of sustained performance across the fleet was the implementation of a structured knowledge-sharing and learning framework that eliminated silos between rigs. With multiple rigs operating simultaneously, the campaign facilitated continuous cross-rig collaboration through scheduled virtual sessions, twice-weekly reviews, and detailed after-action analyses at the end of each well. When one rig achieved a breakthrough—such as improved tripping practices or enhanced bit designs—the lessons were promptly shared across the fleet. This accelerated best practice adoption, minimized variability, and ensured scalable improvements.

To support this collaboration, a responsive technical support structure provided real-time advisory input when rigs faced operational challenges. This helped prevent isolated issues from impacting other rigs. A centralized digital knowledge repository captured and distributed learning efficiently. These combined efforts institutionalized continuous learning, sped up innovation adoption, and established a self-sustaining improvement cycle across the fleet.

Safety as a Driver of High Performance. The transformation journey demonstrated that safety and performance are not competing priorities, but complementary outcomes when systematically managed and culturally aligned. Safety remained the foundation of all performance initiatives. ADNOC's safety system was rigorously applied, including detailed risk assessments, stop-work authority (SWA) for all personnel, and structured learning from high-potential near-miss events.

A strong safety culture was actively maintained through daily rig-based HSE talks, IWAP meetings, pre-job safety briefings, ongoing awareness campaigns, and consistent leadership visibility. Weekly fleet-wide HSE calls enabled prompt sharing of incidents and preventive actions. Safety capabilities were further developed through continuous coaching and a structured training matrix, ensuring consistency across all rigs. Daily site tours by rig leadership reinforced safe behavior and front-line accountability.

In parallel, CCTV systems enabled real-time oversight and post-job learning. Footage of both exemplary and unsafe behaviors was shared across rigs to strengthen collective awareness. These efforts empowered every crew member to proactively intervene. As a result, HSE performance improved by 5% quarterly, based on ADNOC's official inspection process—even under the intensified pace of operations.

Technical Approach

As one of the three foundational pillars of the campaign—alongside Organizational Performance Governance and operational execution—the technical strategy served as the engineering backbone of the campaign, enabling high-speed, high-consistency drilling without compromising well integrity. Through integrated design of bits, BHAs, and formation-specific parameter roadmaps—refined through continuous feedback loops—the fleet achieved repeatable performance across varied subsurface conditions. This approach was reinforced by tailored drilling fluid systems, borehole quality controls, and digitally enabled execution supported by predictive analytics and centralized oversight. These elements collectively transformed geological complexity into consistent, record-breaking results.

Integrated Design: Bit, BHA, and Parameter Roadmaps. To enable high-rate, low-risk drilling in the UAE's complex unconventional formations, a unified technical strategy was deployed that combined bit optimization, BHA engineering, parameter roadmaps, and integration of specialized technologies. The bottom-hole assembly (BHA) was meticulously designed through pre-run torque and drag simulations and vibration modeling to ensure directional control, maximize stability, and suppress vibration-related dysfunctions. Configurations included strategically placed stabilizers, near-bit reamers, and use of torque-resistant connections to improve weight transfer and reduce stick-slip and whirl. Bit selection was treated as a dynamic process grounded in offset analysis, rock strength profiling, azimuthal behavior insights [1] and dull grading. PDC bits were customized with optimized cutter density, thermal resistance, and backrake geometry to improve durability in interbedded and abrasive formations. These designs were continuously refined through vendor collaboration and field feedback, improving footage per bit and reducing shock and vibration.

Supporting this was a phase-specific roadmap for drilling parameters. Initial WOB, RPM, and flow rate guidelines—derived from offset data and MSE modeling—were dynamically adjusted using real-time surface and downhole feedback. This enabled predictive steering, balancing performance with tool life and borehole quality.

Advanced technologies enhanced this system. High-speed telemetry minimized survey time and improved trajectory control, directly increasing ROP and reducing W2W durations. Vibration dampening tools improved BHA stability in lateral runs. Motorized RSS tools, though tested, were phased out due to performance variability—reinforcing that technology adoption was based on field-fit, not trend. This integrated approach created a resilient, repeatable foundation for high-speed drilling.

Drilling Fluid Design and Borehole Quality Optimization. To ensure safe drilling and preserve borehole integrity across varied inclinations, a phase-specific fluid strategy was adopted alongside disciplined wellbore conditioning. A tailored water-based mud (WBM) system was applied in vertical intervals (up to ~30°), providing stability and inhibition while meeting environmental standards. Once formation integrity was verified via shoe bond tests (SBT) and no losses were expected, the operation shifted to oil-based mud (OBM) for curve and lateral sections, enhancing hole stability, lubricity, and cuttings transport while preserving overbalance margins.

Drilling efficiency in lateral section was further improved by carefully reducing mud weights throughout the campaign, based on updated pore pressure estimates and a field-calibrated Mechanical Earth Model (MEM). This enabled higher ROP without compromising well control or wellbore stability.

Borehole quality was sustained through strict circulation protocols, mud conditioning, and active crew engagement. Extended circulation before each trip ensured hole cleanliness and minimized pack-off risk. In the laterals, azimuthal gamma-ray tools aided trajectory control and wellbore placement. Rig leaders held daily briefings on ADNOC's nine Stuck Pipe Prevention Rules (SP2R), reinforcing proactive practices. These efforts, aligned with the company's standard stuck pipe protocol, minimized downhole incidents—even with longer laterals. Altogether, disciplined fluid management and geomechanical insight underpinned consistent casing runs and trouble-free completions.

Digital Drilling and RTOC. Digital technology played a central role in the campaign—not as an add-on, but as an embedded enabler of consistency, reliability, and learning across the fleet. Rig operations were digitally augmented through advisory systems that translated offset well data, formation-specific guidelines, and engineering inputs into real-time parameter windows. These systems didn't just suggest best practices—they tracked compliance, provided feedback to drillers, and introduced a gamified performance element that actively engaged the rig crew. By doing so, each rig crew became a direct stakeholder in continuous improvement. Artificial intelligence and machine learning tools - already proven in the drilling industry [2], [3] - further enhanced the system. Predictive models were used to anticipate drilling dysfunctions, fine-tune parameter windows, and generate real-time recommendations. This intelligence helped shift the operating model from reactive correction to proactive steering. While one AI advisory system provider was used across some rigs, the success came not from the technology alone, but from how it was implemented—supporting a disciplined, transparent, and replicable execution model.

To support field operations, a 24/7 Real-Time Operations Center (RTOC) provided consistent monitoring of surface and downhole parameters. Through proactive tracking of torque, drag and hydraulic trends, the RTOC offered an additional layer of oversight. This allowed early identification of potential hazards like poor hole cleaning or deteriorating drilling efficiency, especially during high-risk zones. Collaboration between on-site engineers and remote ones enabled faster decision-making and helped mitigate down time across multiple wells.

Additionally, the daily performance review process became a high-impact routine. Dashboards tracked key KPIs, variance from optimized windows, and opportunities for replication across rigs. This created a closed-loop system that institutionalized lessons learned and transferred them between pads and crews—accelerating the campaign's learning curve.

This digital-first strategy proved that high performance was not just a function of speed, but of structured insight, timely feedback, and crew engagement powered by intelligent systems.

Operational Rhythm and Well Delivery Optimization

The operational pillar formed the decisive edge of the campaign, turning design intent into execution certainty. At its core was a relentless focus on frontline rhythm, time control, and process discipline. Frontline leaders internalized performance targets through real-time metrics, standardized procedures, and structured coaching—executing thousands of connections, hundreds of trips, and skids with repeatable precision. Cost and flat time were systematically targeted through offline activity planning, invisible lost time reduction, and just-in-time logistics. These operational gains didn't just enable speed—they institutionalized it, making high performance the new normal. The following subsections describe how executional discipline translated into measurable impact across the well delivery cycle.

Drilling Efficiency Optimization. Drilling efficiency was a cornerstone of the transformation campaign, achieved through a deliberate strategy of aggressive yet controlled optimization across all operational elements. Rather than relying on a single breakthrough, performance gains were delivered through the disciplined enhancement of drilling parameters, connection practices, invisible lost time (ILT) targeting, and rig-site KPI management. These improvements established a new baseline of operational excellence across the UC fleet.

At the core of this effort was a customized roadmap for drilling parameters, developed and continuously refined for each formation based on historical well performance and area experience. This roadmap guided the safe and efficient adjustment of weight-on-bit, rotary speed, flow rate and mud properties. Weak formations were specifically accounted for to avoid common issues like losses. Hole cleaning was continuously observed on three levels, real-time monitoring of downhole torque and drag [4], coupled with frequent hydraulic simulations and shale shaker condition surveillance, ensuring that higher ROP did not come at the expense of borehole quality.

Record ROPs were achieved as a result of this optimized approach, all while maintaining operation reliability and avoiding hole problems. Connection time was similarly targeted through standardized procedures, shift-based training, and careful elimination of non-value steps such as unnecessary wash or ream cycles. Time-motion studies helped crews consistently execute best practices. These efforts yielded up to 60% reduction in weight-to-weight (W2W) connection times. This not only saved time cumulatively over tens of thousands of connections through the campaign, but also improved hole quality and reduced the risk of stuck pipe events.

In parallel, Invisible Lost Time (ILT)—the inefficiencies hidden within so-called "normal" operations—was systematically uncovered and eliminated. By comparing real-time execution data against technical limits for each rig activity, teams identified latent inefficiencies in tripping speed, circulation readiness, pipe handling, and inter-crew handovers. These were tackled using LEAN principles, root cause analysis, and cross-rig performance benchmarking. Unlike traditional NPT, ILT required deliberate visibility and management—and this campaign treated it as a core operational focus, enabling significant gains in execution consistency and total well cycle time.

Underlying all of these improvements was a disciplined KPI monitoring and feedback system embedded into the daily drilling rhythm. Rigs tracked key metrics—ROP, connection time, flat time, and NPT—updated twice daily for benchmarking and performance review. This real-time visibility enabled same-shift corrections, immediate accountability, and positive competition. Over time, the consistent focus on execution metrics established a predictable, steady, and performance-driven rhythm across the fleet, where every crew understood the plan, pace, and their role—resulting in a faster, smarter, and more synchronized drilling operation.

Critical Path Management and Offline Time Optimization. Reducing well delivery time required more than faster drilling—it demanded a full reengineering of operational sequencing and task execution under complex downhole conditions. A key breakthrough involved minimizing rig idle time by shifting activities off the critical path and maximizing parallel execution. The team adopted a proactive approach grounded in readiness, coordination, and structured offline planning. Activities such as wellhead installation, casing doubles make-up, float equipment preparation, and cased hole logging were conducted in parallel with ongoing drilling phases.

Daily planning, pre-job checklists, and float plans ensured service providers were aligned and mobilized on schedule. The use of Fluted Mandrel Hanger wellhead systems eliminated wait-on-cement time, saving up to 20 hours per well, while offline logging contributed additional 12 hours of savings.

Additionally, batch drilling played a major role in time optimization. Initially deployed section-by-section, the strategy evolved to commingle multiple sections per skid, enhancing resource efficiency and time utilization. Each rig also developed a standardized skidding SOP, ensuring safe and efficient skidding between wells.

Collectively, these innovations significantly cut flat time and accelerated well delivery. The campaign shifted from a "execute when ready" mindset to "plan ahead and execute in parallel." —enabling safer, faster, and higher-quality drilling outcomes.

Operational Reliability and Readiness. To sustain the accelerated drilling tempo achieved during the campaign, a parallel transformation was undertaken across supply chain, logistics, and maintenance systems. The mission was clear: eliminate operational bottlenecks originating from equipment readiness, material delays, and tool reliability.

Static supply models were replaced with dynamic inventory optimization, balancing cost and availability. A vendor-managed inventory (VMI) program was introduced for consumables, enabling on-demand supply while shifting holding costs to vendors. These changes drove supply-related NPT near zero, even in remote locations.

Maintenance also evolved from reactive repair to predictive reliability. Condition-based protocols were rolled out, using real-time vibration, fluid, and temperature data to monitor equipment health. Interventions were planned during idle periods, with parts kitting and pre-assembly minimizing repair times. Recognizing harsh downhole conditions, a strict qualification processes was mandated, especially for directional service providers—leading to measurable increases in MTBF and service consistency.

These improvements in logistics and maintenance quietly but powerfully enabled the campaign's success. By ensuring the right tools were always available—and that they worked reliably—the team safeguarded operational continuity. This integration of supply precision and proactive reliability became a silent yet foundational enabler behind the campaign's ability to drill faster, safer, and more consistently.

Results and Performance Outcomes

This section presents a comprehensive analysis of the campaign's performance outcomes, spanning drilling efficiency, cost optimization, safety, and environmental impact. By systematically tracking key indicators such as Rate of Penetration (ROP), connection time (W2W), Non-Productive Time (NPT), well delivery duration, and cost per foot—alongside HSE performance and carbon footprint—this section quantifies the tangible impact of ADNOC's unconventional drilling transformation. The results reflect not only technical excellence, but also operational discipline, cultural maturity, and strategic alignment with broader corporate goals.

Rate of Penetration

The campaign established a new performance frontier in the UAE's UC operations. Across all hole sizes, extensive rig-site follow-up, sustained parameter optimization, BHA refinement, and strategic bit selection led to unprecedented ROP achievements. Peak ROP values reached 178.3 ft/hr in the 17-1/2″ section, 137.4 ft/hr in the 12-1/4″, and 124.3 ft/hr in the 8-1/2″ sections—performance levels never previously achieved in these complex formations. These were not isolated events but the outcome of a systematic and repeatable optimization strategy.

When comparing campaign phases, the improvement was even more remarkable: between H2-2023 and H1-2025, average ROP increased by 203% in the 17-1/2″ section, 141% in the 12-1/4″ section, and 211% in the 8-1/2″ section. These trends confirm a steep and well-managed learning curve.

Scatter plots of ROP over time for all three hole sections [Fig-1] further validate this performance leap. The 17-1/2″ section showed a notable upward trend with $R^2 = 0.406$, while both the 12-1/4″ and 8-1/2″ sections exhibited strong positive correlations with R^2 values exceeding 0.51. Notably, the consistent slope of ~0.12–0.14 ft/hr per day across all three sections confirms that performance uplift was not only measurable, but also systematic and replicable across the unconventional fleet.

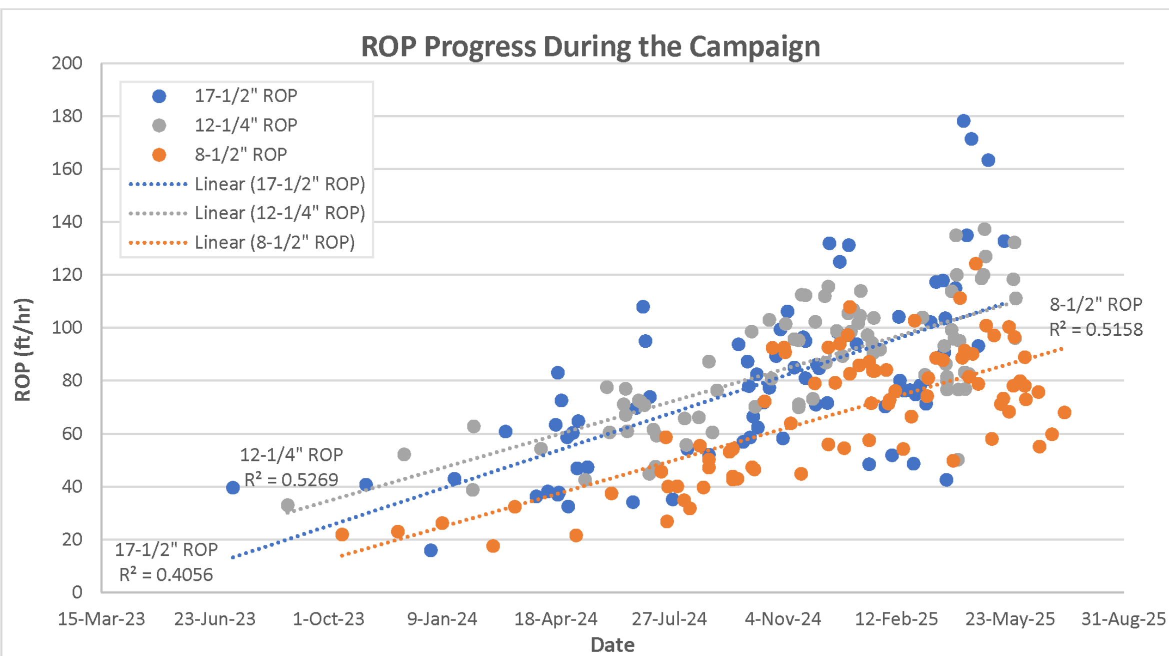

Figure 1—Campaign-wide improvement in Rate of Penetration (ROP)

Operationally, the fleet set a new benchmark by drilling 3,580 feet in a single 24-hour period, the highest daily footage so far. These gains were not coincidental. They reflect a disciplined technical approach where aggressive yet controlled parameters, vibration-suppressing BHAs, and adaptive field practices translated speed into sustainable, consistent performance—well after well.

Importantly, this momentum aligns directly with ADNOC Onshore's strategic ambition. Inspired by the vision set by the CEO, the UC drilling organization has embraced the goal of delivering one mile per day (5,280 ft)—a milestone that was once aspirational but is now approaching our operational reach. The records achieved during this campaign represent foundational progress toward that target, proving that world-class drilling efficiency is both achievable and replicable across the entire fleet.

Connection Time Optimization: Precision in the Margins

While drilling performance gains often center around ROP improvements, this campaign demonstrated that record-setting well delivery is equally driven by optimizing time between connections—what is often considered "non-drilling" time. By focusing on Weight-to-Weight (W2W) connection performance, the fleet unlocked significant efficiency without compromising borehole quality or safety.

Between Q3-2024 and Q2-2025, average Weight-to-Weight (W2W) connection time was reduced by 60% in the 17-1/2″ section, 53% in the 12-1/4″ section, and 57% in the 8-1/2″ section, reflecting consistent efficiency gains across the entire wellbore.

This translated to a combined saving of over 1.25 rig days per well—purely from connection time improvements. Given the scale of the campaign, this represents hundreds of days returned to the asset base, underscoring that performance uplift was not limited to drilling speed but extended to process efficiency at every level.

Trendline analysis [Fig-2] confirmed the consistency and strength of these improvements, with R^2 values around 0.6 across all sections—indicating strong negative correlations between W2W duration and the campaign timeline. Exponential regression models showed even better fit, suggesting that improvement rates accelerated as the campaign progressed. This trend reflects more than just technical enforcement; it signals a fleet-wide cultural shift, where crews internalized faster, more efficient connection practices as a new operational norm.

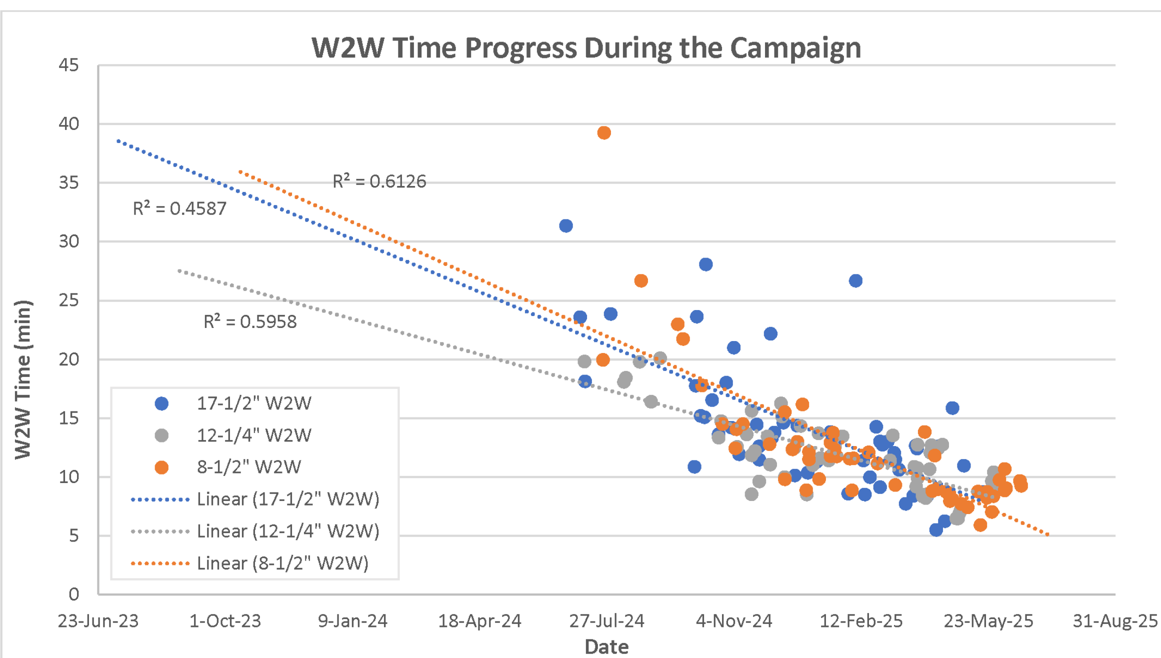

Figure 2—Reduction in Weight-to-Weight (W2W) connection time across the campaign

Crucially, this transformation was not automatic. At the outset, many rig leaders were hesitant to accelerate connections due to perceived risks—particularly in horizontal intervals. However, visible leadership support, structured reinforcement, and real-time communication of results gave the entire team the confidence to adapt. What was once seen as a risk had became a shared win. The initiative became a case study in how empowerment, data, and coaching can drive lasting change.

This section underscores that ADNOC's drilling success was not only about "drilling faster" but also about working smarter across the entire well cycle. The commitment to optimizing every operational detail redefined what was possible for well delivery. It reflects a culture of high performance where leadership, data, and frontline discipline converged to turn non-drilling time into competitive advantage.

Learning Under Load: What One Rig Taught Us About Controlling NPT

In high-performance drilling environments, pushing the operational envelope often brings an inherent rise in technical risk. This trend was clearly illustrated by Rig-A [Fig-3], which served as a representative case study within the campaign. As the rig progressed through three successive pads with increasingly aggressive drilling targets, its NPT profile evolved accordingly: 7.22% in Pad-1, rising to 17.17% in Pad-2, before being brought back down to 11.48% in Pad-3.

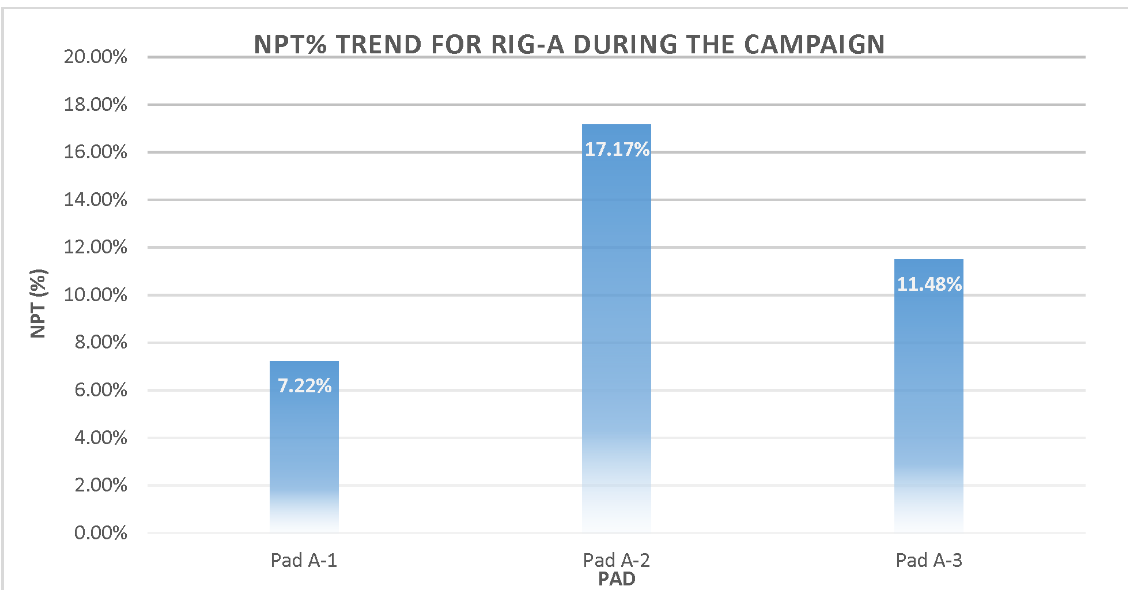

Figure 3—Evolution of Non-Productive Time (NPT) profile on a representative rig

Rather than interpreting this rise as regression, the team viewed it as a predictable side effect of pushing system limits—a reflection of the fact that "no work means no problems, but great work often invites greater challenges." Crucially, the team didn't accept higher NPT as a permanent trade-off for speed. Instead, they responded with intensified engineering oversight, upgraded tool selections, and closer contractor integration, which successfully reversed the upward trend and restored control.

This pad-level analysis demonstrates not only transparency in performance monitoring but also the fleet's growing resilience—learning to absorb complexity while maintaining forward momentum. As the campaign matured, these learnings were transferred across rigs, embedding a more proactive and responsive NPT management culture fleet-wide.

While the numbers reflect the growing pains of a high-tempo campaign, they also affirm the system's maturity: challenges were addressed through structured root-cause analysis and collaborative action. This phase underscored that transparency, combined with technical engagement, is critical to sustaining momentum in complex operations—and laid the groundwork for measurable NPT reduction in the next campaign cycle.

Well Delivery Acceleration: The First Tangible Outcome of Systematic Optimization

The campaign's well delivery trajectory reflects a strategically managed performance transformation—marked by structured acceleration, operational discipline, and continuous refinement. Average well delivery time dropped from 96.5 days in Q3-2023 to 18.95 days in Q2-2025, representing a reduction of nearly 80% across eight consecutive quarters [Fig-4]. This performance evolution exhibits a strong logarithmic pattern ($R^2 = 0.9854$), illustrating how efficiency gains were increasingly institutionalized through data-driven execution, not episodic wins.

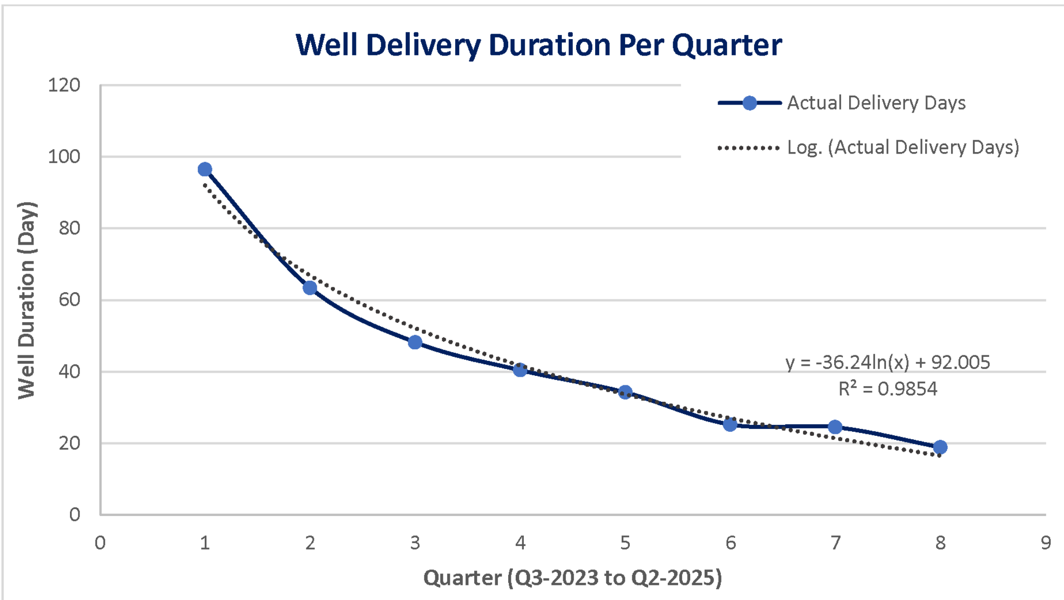

Figure 4—Reduction in average well delivery time across the campaign

Rather than isolated gains, this sustained improvement was the product of deliberate interventions: compressing connection times, minimizing invisible lost time, enforcing parameter roadmaps, and standardizing high-performance BHAs and bit programs. The current average of 18.95 days per well demonstrates that ADNOC's unconventional fleet has entered a high-efficiency zone, where operational excellence is no longer theoretical—it is measured, repeated, and scalable.

Critically, the campaign penetrated the 15-day barrier and has already strongly approached 14-day performance—evidence that the system is not only optimized but still evolving. While each successive gain may require sharper focus, the trend affirms that improvement remains both possible and underway. This well delivery performance not only elevates ADNOC Onshore's operational benchmarks but also provides a replicable blueprint for unconventional field development across the region and beyond.

Redefining Cost-Efficient Drilling

The campaign achieved a steep and sustained reduction in well construction costs, with average cost per foot decreasing from 668 USD/ft in H2-2023 to just 430 USD/ft by H1-2025 [Fig-5]. Logarithmic regression of the consolidated trend yielded a strong correlation ($R^2 = 0.951$), underscoring that cost optimization was not incidental but structurally embedded into the drilling system. Notably, several recent wells were drilled for nearly 300 USD/ft, reflecting best-in-class execution.

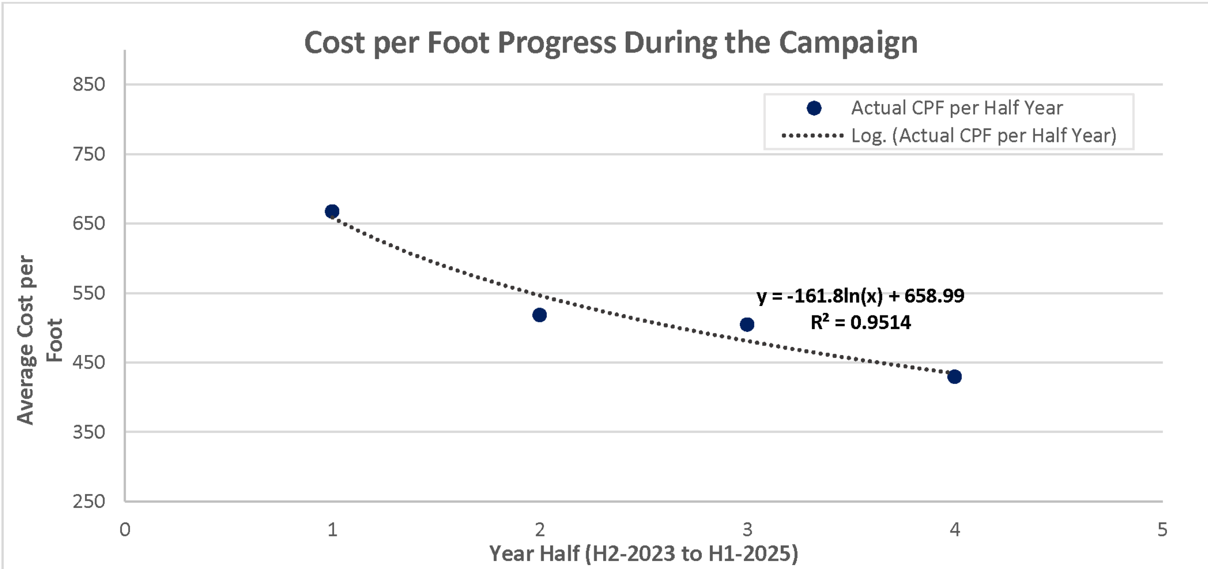

Figure 5—Cost per foot reduction achieved across the UC drilling campaign

This reduction directly enhances the economic viability of unconventional resource development in the UAE, transforming once-marginal prospects into commercially attractive drilling targets. By minimizing both productive time and NPT, the campaign unlocked cost savings that lowered the breakeven point for field development and freed capital for reinvestment into additional wells.

Beyond individual well economics, the accelerated drilling pace enabled faster resource monetization, bringing production online ahead of the original development schedule. As a result, the inventory of commercially viable drilling locations has expanded, significantly increasing both the recoverable resource base and the long-term production potential of ADNOC Onshore's UC portfolio.

HSE and Well Control Outcomes

The campaign achieved exceptional safety and well control outcomes, validating that performance acceleration does not come at the expense of operational integrity. Across the entire campaign period, zero Lost Time Incidents (LTIs) and zero well control events were recorded—despite the aggressive well delivery timelines. This underscores the maturity of the fleet's HSE culture and the effectiveness of pre-job planning, crew engagement, and active risk management.

Independent HSE audits conducted quarterly by ADNOC Onshore further reinforce this track record, with the fleet's audit scores consistently improving over the course of the campaign [Fig-6]. These continuous improvements reflect the disciplined execution of internal punch list campaigns, daily 1-to-1 HSE coaching sessions, and a culture of accountability that empowered all crew members to act on Stop Work Authority (SWA) whenever necessary.

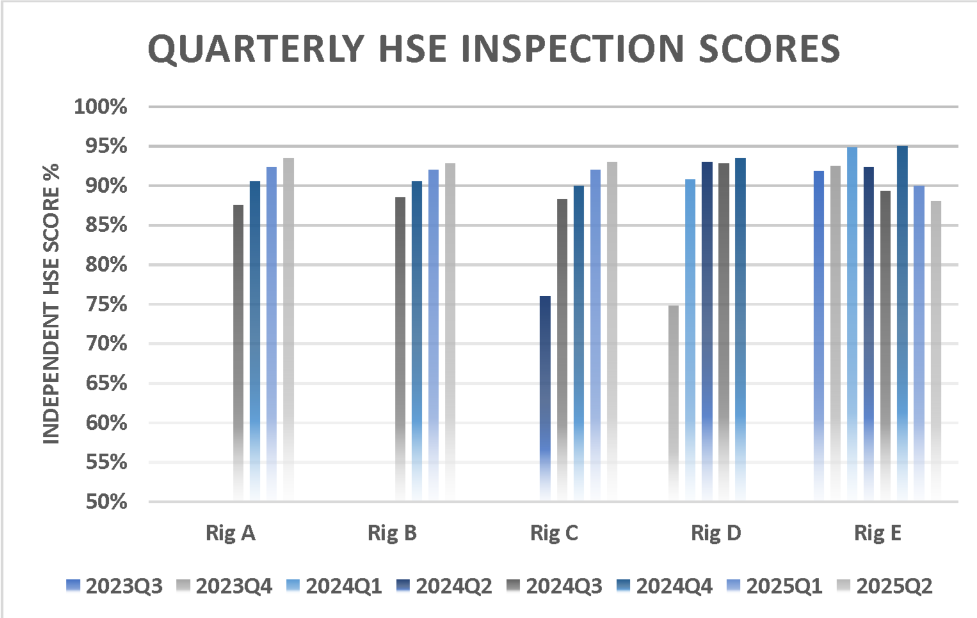

Figure 6—Trend in independent HSE audit scores for multiple rigs

Notably, this outstanding HSE record was achieved under technically complex conditions, where the margin for error was inherently narrow. Maintaining primary well control throughout all phases of drilling and completion demonstrates the robustness of the operational envelope and the team's ability to execute under high-performance conditions without compromising safety or quality.

This convergence of speed, safety, and well control illustrates that ADNOC's drive for efficiency is firmly rooted in a risk-managed, systematized drilling framework—a model that can be replicated across the unconventional portfolio.

Sustainability and CO_2 Emissions

The environmental benefits realized from this drilling campaign are not isolated achievements, but integral to ADNOC's overarching environmental, social, and governance (ESG) goals. A deep-dive analysis on a single rig within the fleet [Fig-7] revealed that active rig-site parameter optimization—combined with BHA optimization—resulted in a 42.3% reduction in diesel consumption per foot drilled. This improvement, validated by a strong linear correlation (R^2 = 0.681), reflects sustained gains in energy efficiency driven by disciplined, performance-focused drilling practices. Based on actual diesel consumption data and the standard emission factor of 2.68 kg CO_2 per liter [5], the rig achieved an estimated reduction of 4,552 metric tons of CO_2 emissions throughout three pads. This is equivalent to removing nearly 1,000 passenger vehicles from the road for a full year, according to EPA benchmarks [6]. These results demonstrate that operational excellence and environmental stewardship are not competing goals, but dual outcomes of intelligent system design and rig site execution. These results position the UC drilling campaign as a strategic pillar in delivering the company's long-term vision for a cleaner, more efficient energy sector.

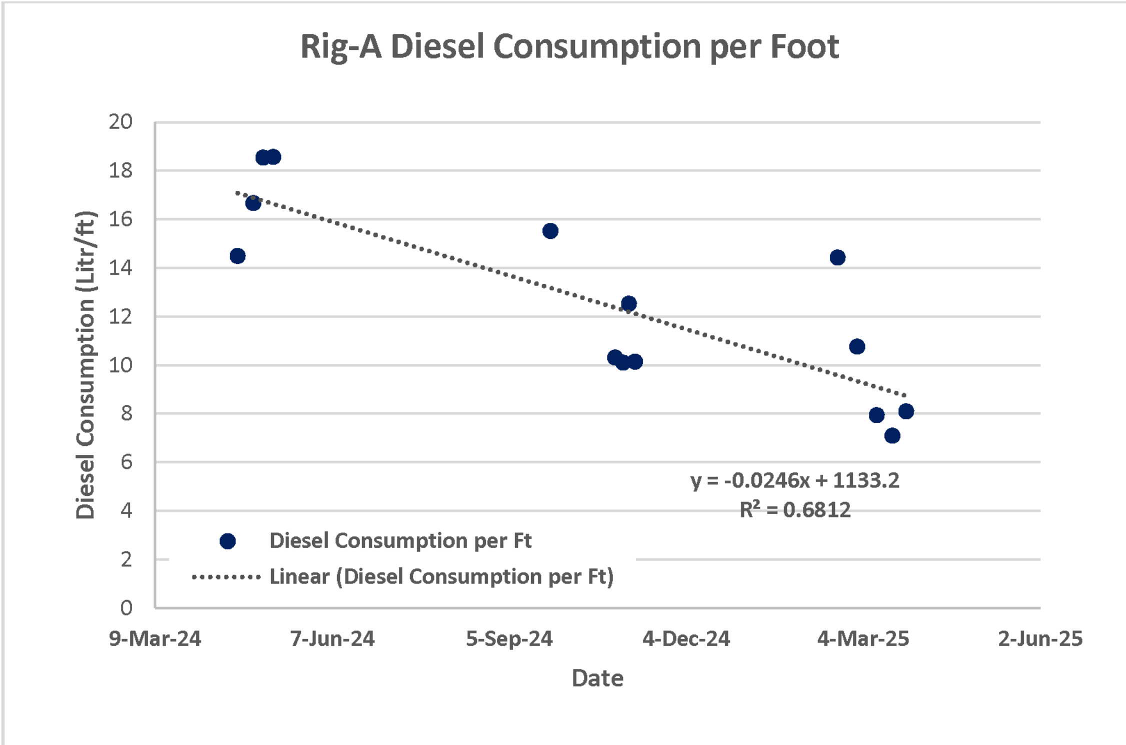

Figure 7—Reduction in diesel consumption per foot drilled

Discussion

The transformation of ADNOC Onshore's unconventional drilling performance was not the result of isolated breakthroughs but the outcome of deeply integrated systems thinking—where leadership, data, technology, and frontline execution aligned seamlessly. The campaign delivered not only operational gains but strategic insights relevant to complex drilling environments globally.

The most powerful enabler was the shift from incremental gains to systems-level transformation. Instead of episodic enhancements or standalone technologies, the campaign established a closed-loop model: real-time rig data informed on-the-fly parameter adjustments, evaluated against technical benchmarks, and institutionalized if successful. Drilling parameter roadmaps evolved from static guides into dynamic tools—adapted daily through KPI reviews. This agility extended beyond operations reflecting a cultural shift toward continuous progress, clear communication, and net impact measurement.

Empowering rig leadership further differentiated the campaign. Day-to-day decisions moved closer to execution, backed by dashboards, coaching, and structured governance. Accountability transitioned from top-down directives to shared commitment. Rig crews became co-owners of performance, proposing refinements, reducing ILT, and embracing targets not as constraints, but as launchpads. This mindset—anchored in responsibility and integration—reframed goals from "meeting plans" to "redefining limits."

Early challenges, especially tool failures in high-temperature zones, were treated as catalysts. Rather than normalizing setbacks, teams intensified tool qualification, enhanced BHA design, and improved contractor alignment. For example, the motorized RSS tool was tested and phased out due to reliability issues—demonstrating a culture where efficiency and suitability overruled popularity. Similarly, rising NPT on aggressive wells triggered preemptive mitigation—eventually reversing trends through predictive interventions and adaptive designs. These actions reflect a system not content with complacency, but one that learns and adjusts in real time.

Data integration was central to this evolution. Predictive analytics, high-speed telemetry, and advisory systems were embedded not as passive monitors, but as behavioral drivers. Compliance with optimized parameter windows was tracked and fed back to rig crews—creating a gamified feedback loop that fueled peer motivation and real-time course correction. The RTOC enabled cross-rig collaboration, providing early warnings and supporting fast, impactful decisions.

The results were substantial. Well delivery time dropped by 80% across the campaign, with multiple wells drilled under 15 days. Connection time improved by up to 60%. Cost per foot decreased by 35.6%, reaching near $300/ft in some wells. ROP records reached 178.3 ft/hr in 17-1/2″, 137.4 in 12-1/4″, and 124.3 in 8-1/2″ sections—backed by strong trendlines and reduced tool failures.

None of these achievements came at the expense of safety or sustainability. The campaign recorded zero well control events and zero LTIs, even under intensified pace. Diesel consumption per foot dropped over 40%, cutting an estimated 4,552 tons of CO_2—equivalent to removing 1,000 cars from the road for a year. These gains directly advanced ADNOC's "Maximum Energy, Minimum Emissions" agenda and embodied ADNOC's 100% HSE commitment —demonstrating that efficiency and sustainability are mutually reinforcing outcomes.

The most enduring legacy may be how the campaign redefined the role of learning. Best practices were codified, replicated, and scaled across pads. The system was designed not just to succeed—but to evolve. Innovation was no longer a one-time effort—it became the standard rhythm of execution. This campaign did not merely break records—it built a transferable model of collaborative, efficient, and high-impact drilling. The result was not only faster wells, but a smarter, safer, and cleaner model for unconventional resource development—built in the UAE, and ready to inspire drilling excellence worldwide.

Conclusions and Recommendations

Conclusions

ADNOC Onshore's UC drilling transformation stands as a benchmark for what is achievable when strategic system design, disciplined execution, and empowered leadership converge. The campaign safely delivered over one million feet while reducing well delivery time by 80% and cost per foot by 35.6%—without compromising safety, borehole quality, or environmental responsibility. These results confirm that performance excellence is not coincidental, but deliberately engineered.

The transformation stemmed from an integrated framework of three synchronized pillars: strategic management provided the governance, culture, and accountability to drive continuous improvement. Technical design—including optimized BHAs, adaptive parameter roadmaps, and disciplined fluid strategies—converted geological complexity into repeatable success. Operational execution brought consistency, speed, and control to the rig site, enabling sustained delivery at scale.

Notably, these gains were achieved under extreme drilling conditions, proving that even harsh environments can yield record-setting results with the right system in place. Moreover, the campaign delivered over 40% reduction in diesel consumption per foot, directly supporting ADNOC's "Maximum Energy, Minimum Emissions" agenda.

More than a project closeout, this transformation established a new baseline for UAE unconventionals and created a scalable, exportable model for global drilling performance improvement. The following key conclusions summarize the campaign's outcomes:

1. The campaign safely delivered over 1 million feet of unconventional wells.
2. Average well delivery time was reduced by 80% across eight consecutive quarters.
3. Drilling cost per foot dropped by 35.6%, enabling expanded development.
4. ROP records were set across all hole sizes, with strong repeatability trends.
5. Zero well control events and zero LTIs validated operational integrity.
6. CO_2 emissions per foot were reduced by over 40%, supporting ADNOC's sustainability agenda.
7. Empowered leadership, not just technical upgrades, proved essential to transformation.

8. A holistic strategy—not isolated improvements—drove systemic, scalable performance.
9. The model demonstrated is applicable across ADNOC and globally in similar contexts.

Recommendations

Drawing from this experience, several actionable recommendations emerge for operators and drilling professionals aiming to replicate similar transformations:

- **Design for System, Not Just Speed**
  Prioritize end-to-end integration across management, engineering, and execution. Isolated optimizations must give way to systemic thinking that aligns all actors around common goals.
- **Empower Frontlines with Accountability and Tools**
  Shift authority closer to the point of execution. Equip rig leaders with real-time KPIs, coaching, and autonomy to act. Performance ownership must be shared to be sustained.
- **Institutionalize Learning Through Knowledge Transfer Systems**
  Build structured platforms for capturing, sharing, and scaling best practices. Use cross-rig knowledge loops to reduce variability and embed excellence.
- **Treat Safety and Sustainability as Performance Drivers**
  Do not trade safety for speed. Reinforce that safe operations and emissions reductions—through optimized rig days and fuel efficiency—are indicators of system maturity, not constraints.
- **Gamify Compliance and Feedback**
  Engage rig crews by transforming data into feedback loops, performance dashboards, and challenge-driven culture. Behavior change accelerates when visibility and motivation are built into execution.

Beyond millions of feet drilled, dollars saved, and tons of CO_2 avoided, the most enduring achievement of this campaign may be cultural. ADNOC Onshore proved that world-class drilling is not defined by hardware or geography—it is defined by people who believe that limits are there to be surpassed, or as ADNOC Onshore's CEO always remind us "*Records are meant to be broken*". This performance legacy is now embedded in the crews, systems, and strategy of a company not just prepared for the future, but determined to lead it.

Acknowledgements

The authors thank ADNOC Onshore for supporting this work and acknowledge the contributions of our leadership, supervisors, engineers, partners, contractors, service providers and drilling crews whose efforts made this performance transformation possible.

Acronym

EPA	Environmental Protection Agency
ILT	Invisible Lost Time
IWAP	Integrated Work Activity Planning
LTI	Lost Time Incident
MEM	Mechanical Earth Model
MSE	Mechanical Specific Energy
MTBF	Mean Time Between Failures
NPT	Non-Productive Time
R^2	Coefficient of Determination
SP2R	Stuck Pipe Prevention Rules
SWA	Stop Work Authority

References

1. M. Nour, M. S. Farahat, and O. Mahmoud, "The Influence of Directional Well Azimuth on Bit Performance: A Statistical Approach," in *Volume 10: Petroleum Technology*, Hamburg, Germany: American Society of Mechanical Engineers, June 2022, p. V010T11A015. doi: 10.1115/OMAE2022-80570.
2. M. Nour, M. S. Farahat, and O. Mahmoud, "Picking the Optimum Directional Drilling Technology (RSS vs PDM): A Machine Learning-Based Model," in *Volume 10: Petroleum Technology*, Hamburg, Germany: American Society of Mechanical Engineers, June 2022, p. V010T11A014. doi: 10.1115/OMAE2022-80569.
3. M. Nour, S. K. Elsayed, and O. Mahmoud, "A supervised machine learning model to select a cost-effective directional drilling tool," *Sci Rep*, vol. **14**, no. 1, p. 26624, Nov. 2024, doi: 10.1038/s41598-024-76910-z.
4. M. Nour, A. A. Elgibaly, M. S. Farahat, and O. Mahmoud, "Novel Torque and Drag Model for Drilling Two-Dimensional High-Angle Wells," *ACS Omega*, vol. **7**, no. 14, pp. 12374–12389, Apr. 2022, doi: 10.1021/acsomega.2c00924.
5. Y. Gilbert, M. Hagström, and J. M. Getliff, "Reducing the Carbon Footprint of Drilling and Completion Operations," in SPE International Conference on Health, Safety and Environment in Oil and Gas Exploration and Production, Rio de Janeiro, Brazil: SPE, Apr. 2010, p. SPE-126756-MS. doi: 10.2118/126756-MS.
6. O. US EPA, "Greenhouse Gas Equivalencies Calculator." Accessed: Aug. 05, 2025. [Online]. Available: https://www.epa.gov/energy/greenhouse-gas-equivalencies-calculator